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SAFE: Sound Analysis for Fall Event detection using machine learning

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ABSTRACT

This study evaluates the application of machine learning (ML) and deep learning (DL) algorithms for fall detection using sound signals. The work is supported by the Sound Analysis for Fall Events (SAFE) dataset, comprising 950 audio samples, including 475 fall events recorded with a grappling dummy to simulate realistic scenarios. Decision tree-based ML algorithms achieved a classification accuracy of 93% at lower sampling rates, indicating that critical features are preserved despite reduced resolution. DL models, using spectrogram-based feature extraction, reached accuracies up to 99%, surpassing traditional ML methods in performance. Linear models also achieved high accuracy (up to 97%) in various spectrogram techniques, emphasizing the separability of audio features. These results establish the viability of sound-based fall detection systems as efficient and accurate solutions.

1. Introduction

In recent years, the occurrence of falls among old people has gained substantial attention as a major public health issue in the United States. As presented in a report by the CDC, unintentional falls have become the leading cause of injury and injury-related deaths among adults 65 years and older in the United States (Kakara, 2023). From 2020 to 2021, there was a marked increase in nonfatal and fatal falls among older adults. Specifically, about 36% of older adults reported experiencing at least one fall and there were 52 fall-related deaths per 100,000 people in this demographic. This trend highlights the critical need for fall prevention and health care strategies to address this important public health problem.

Extensive research has focused on developing systems and algorithms to enhance the functional abilities of elderly people, leading to the evolution of three primary fall detection techniques (Mubashir, Shao, & Seed, 2013). Wearable device-based methods in fall detection incorporate sensors such as accelerometers (Hussain, Hussain, Ehatisham-ul Haq, & Azam, 2019; Santos et al., 2019), often integrated into smartphones (Hassan, Gumaei, Aloi, Fortino, & Zhou, 2019), to analyze movement, posture (Feng, Yu, Naqvi, & Chambers, 2014) and audio features (Cheffena, 2016). Ambient device-based approaches use environmental sensors, including acoustic (Adnan et al., 2018; Droghini, Ferretti, Principi, Squartini, & Piazza, 2017; Droghini, Principi, Squartini, and Olivetti & Piazza, 2018; Droghini, Squartini, Principi, Gabrielli, & Piazza, 2021; Li, Ho, & Popescu, 2012, 2014; Popescu, Li, Skubic, & Rantz, 2008; Principi, Droghini, Squartini, Olivetti, & Piazza, 2016; Zhuang, Huang, Potamianos, & Hasegawa-Johnson, 2009) and pressure sensors (Litvak, Zigel, & Gannot, 2008), to detect changes in the environment. Vision-based techniques, on the other hand, use cameras and computer vision algorithms to analyze visual data for irregular movements or postures, offering a complete perspective on fall detection (Asif et al., 2020; De Miguel, Brunete, Hernando, & Gambao, 2017; Panahi & Ghods, 2018; Töreyin, Dedeo˘ glu, & Çetin, 2005).

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Here is the revised text with the requested formatting:

In a previous study (Huang & Garcia, 2023), a fall detection system was integrated into an orthopedic walker, combining wearable and ambient detection methods. Motion data was gathered using an inertial measurement unit (IMU) sensor, which was analyzed through machine learning for real-time fall detection and activity logging. This integration into a walker enabled continuous monitoring without the need for additional wearable devices, providing a practical solution for elderly care. However, limitations were identified due to the system's dependency on the walker's movement. If the walker remained stationary, falls could not be detected by the IMU sensor. To address these limitations, the augmentation of sound-based fall detection has been explored in this work, alongside the IMU sensor affixed to the walker as detailed in Huang and Garcia (2023). This enhancement is intended to improve the functionality of the system by combining IMU data with audio-based fall detection using machine learning and deep learning approaches.

The following sections of this article will detail the implementation and evaluation of the proposed sound-based fall detection algorithm. Section 2 reviews relevant studies and challenges in collecting fall detection data, noting the lack of publicly available datasets. Section 3 describes the creation of the SAFE dataset, including recording methods, data augmentation, and segmentation into fall and non-fall samples. Section 4 explains the use of machine learning and deep learning models for detection, focusing on feature extraction methods like spectrograms and digital audio processing. Section 5 presents the evaluation results, comparing the model performance between different representations and sampling rates. Section 6 focuses on practical applications, highlighting low-power, privacy-friendly embedded systems. Section 7 addresses study limitations, including dataset constraints and model deployment challenges. Section 8 concludes with findings and proposes future work, such as hardware integration, computational optimization, and improved audio classification methods.

2. Related work

Research on audio classification of human falls is constrained by significant data collection challenges. Acquiring such data is problematic due to the impracticality and safety concerns associated with recording actual human falls. Simulated falls, performed with mats or special equipment, often fail to authentically replicate the sound of real human falls. Additionally, while the use of test dummies is a safer and more accepted method, it does not accurately capture all the acoustic properties of a real human fall.

This situation leads to a considerable lack of available data, requiring the creation of unique datasets for new research in this area. In addition, these specially designed datasets are rarely released to the public, which exacerbates the issue of data scarcity in the field of audio-based human fall detection.

In the research presented by Li et al. (2012), a fall detection system is described, using a circular microphone array made up of eight omnidirectional microphones positioned along a circular perimeter. This system is designed to discern the unique acoustic signatures of human falls within an ambient auditory environment. It achieves sound source localization with high precision by employing the steered response power with the phase-transform technique, which is adept at mitigating the distortions caused by background noise and echoes. Following the localization phase, the system employs beamforming to refine the quality of the audio signal, and integrates the height from which the sound originates to increase its detection accuracy. The classification task is executed by a Nearest-Neighbor classifier. This classifier utilizes the spectral features extracted via mel-frequency cepstral coefficients from the enhanced audio signal. To validate the operational effectiveness of this system, a specialized dataset was created. This dataset, which was not released to the public, included 120 positive (fall) and 120 negative (no-fall) samples. These samples were recorded in a controlled laboratory environment, with stunt actors dropping onto a mat. The study achieved a peak accuracy of 98% in the classification of fall and non-fall events.

In the research presented in Principi et al. (2016), the authors describe a fall detection system that combines a floor acoustic sensor with an overhead microphone. This system employs a Support Vector Machine algorithm that leverages Mel-Frequency Cepstral Coefficients for sound classification. The efficacy of this algorithm was tested using a dataset of 308 samples, categorized into seven distinct classes. These samples varied in distance and ambient noise levels to simulate realistic scenarios.

For replicating human fall sounds, the study used a "Rescue Randy" dummy, which is frequently used in similar research (Alwan et al., 2006; Litvak et al., 2008; Zigel, Litvak, & Gannot*, 2009). The algorithm achieved an average accuracy rate of 98%, with remarkable proficiency in detecting falls involving the "doll" class (representing the dummy), achieving a detection rate of 100% in certain instances. However, it should be noted that this dataset is no longer available online.

In Cheffena (2016), the development of an automated fall detection system using smartphone audio features is presented. It describes the use of techniques for feature extraction, including spectrogram, Mel frequency cepstral coefficients, linear predictive coding, and matching pursuit. The study explores four distinct machine learning classifiers: the k-nearest neighbor classifier, support vector machine, least squares method, and artificial neural network for their efficacy in differentiating between fall and non-fall events. Evaluation of each classifier's performance is conducted using metrics such as sensitivity, specificity, accuracy, and computational complexity for each audio feature. In particular, the highest performance is observed when using spectrogram features in conjunction with the neural network classifier, achieving over 98% in accuracy. The study used a dataset that was gathered using smartphones and included the participation of nine volunteers. This dataset included a broad spectrum of fall scenarios, incorporating instances of falls occurring during sleep, while seated, and from a standing position, with a total of 130 cases. Additionally, the researchers recorded 26 distinct non-fall actions. This resulted in the generation of 130 samples for the no-fall category, bringing the entire dataset to a total of 260 samples. However, the final dataset compiled for this study has not been made publicly available by the authors.

In Kaur, Wang, and Shi (2022) authors used Audio Transformers to detect falls in bathroom environments, avoiding the need for wearable devices. They generated a new dataset from 12 volunteers, aged 15 to 50, who acted out 8 bathroom fall scenarios. This dataset, designed to improve model accuracy and minimize false positives, consisted of 82 unique audio recordings made using smartphones. Through data augmentation, the dataset grew from 82 to 4390 instances, including 855 falls and 3535 non-falls. The researchers achieved a maximum accuracy of 86.7% by testing various transformer configurations. They transformed the audio samples into image formats using log-Mel spectrograms for detailed analysis. Due to confidentiality, the study data remains inaccessible to the public. The team evaluated the effectiveness of the fall detection algorithm using various machine learning models and feature sets, including SVM, DNN, and Transformer-based models, and different representations of audio signals.

Other studies such as Droghini et al. (2021), Droghini, Vesperini, Principi, and Squartini and Piazza (2018), Geertsema, Visser, Viergever, and Kalitzin (2019), Li et al. (2014), Töreyin et al. (2005) and Zhuang et al. (2009) have similarly explored the use of multiple Machine Learning techniques for audio classification to identify falls. However, none of these studies provides publicly available datasets. Although some datasets were initially published online, they are no longer accessible today.

In summary, audio-based fall detection has been the focus of several research studies; however, these efforts have been characterized by recurring challenges and limitations that reveal significant gaps in the field:

- Dataset availability has been a major challenge in this domain due to the lack of publicly available datasets. This limitation has forced researchers to develop independent datasets, which are often inaccessible to others.
- Dataset development remains a fragmented process, as most studies create their own datasets due to the scarcity of shared resources. This has contributed to limited data accessibility for broader research.
- Data accessibility continues to be an issue. Despite claims by some researchers to make their datasets public, many remain inaccessible, making it difficult to replicate studies and advance the field.
- The use of dummies, such as the "Rescue Randy", to simulate human falls has been common. However, these setups often fail to capture the diversity and complexity of real-world fall scenarios and environmental conditions.
- Machine learning techniques applied in this field have ranged from simple algorithms like k-Nearest Neighbors (kNN) to advanced models like audio transformers. However, comparative evaluations of these techniques within consistent experimental setups have been limited.
- Feature extraction methods predominantly rely on specific techniques, such as Mel-frequency cepstral coefficients (MFCC) and Mel spectrograms. A systematic exploration of alternative feature extraction methods and their influence on classification performance has been lacking.

To address these gaps, this article made the following efforts:

- The SAFE (Sound Analysis for Fall Events) dataset was introduced as a publicly accessible resource containing 950 audio samples, including 475 fall events recorded under various conditions. This dataset was designed to support replicability and facilitate future research.
- Realistic fall scenarios were captured in the SAFE dataset using a grappling dummy in various environments. This effort aimed to reflect a range of fall and non-fall events, improving consistency and standardization across studies.
- The SAFE dataset was released with detailed metadata and documentation to enhance transparency and reproducibility. This approach facilitates replication and ensures trustworthiness in audio-based fall detection research.
- A variety of fall scenarios were simulated using a grappling dummy, including falls on different surfaces and interactions with objects. This approach captured a wider range of acoustic properties and improved the realism of the dataset.
- A range of machine learning and deep learning models were evaluated, including decision trees, support vector machines, and state-of-the-art architectures such as ConvNeXt and ResNet. Their performance in a unified dataset provided information on their relative effectiveness.
- Multiple feature extraction techniques, including Mel spectrograms, STFT spectrograms, and constant Q-transform, were assessed. The analysis focused on identifying the strengths and limitations of each method to guide the selection of appropriate techniques for audio-based fall detection.

These efforts addressed identified gaps and contributed to advancing the field of audio-based fall detection research. The findings serve as a basis for further improvements and practical implementations.

3. Dataset collection

In this section, the process of building the sound classification dataset for fall events is described. The dataset is composed of two classes: positive samples, named "falls", and negative samples, which represent scenarios in which "no falls" occurred.

3.1. Data recording

Initially, multiple approaches were considered to collect fall-related audio data. These included methods such as people falling on a mat or wearing protective gear. However, these approaches were quickly ruled out after preliminary tests indicated a potential risk of injury. Consequently, the focus was on using publicly available fall audio samples from sources such as YouTube or websites that offer free audio samples. However, it was considered difficult to justify this approach due to the unknown origins of the audio.

Fig. 1. Grappling dummy used for recording fall audio samples. It weights about 140 pounds and has a height of 1.80 meter.

Table 1
Characteristics of fall audio data samples in each of the environments where samples were recorded.

ID	Location	Surface	Position	Samples	Surface description
01	Basement	Carpet over concrete	Lying	59	Basement with firm concrete base covered with a layer of carpet.
02	Basement		Standing	126	
03	Lab	Carpet over wood	Lying	26	Upstairs laboratory floor with wooden base covered with a layer of carpet.
04	Lab		Standing	104	
05	Stairs	Concrete	Standing	79	Concrete stairs. The set includes three steps.
06	Wood Platform	Wood	Lying	26	A purely wooden surface, featuring a ramp leading to a flat area.
07	Wood Platform		Standing	55	

Later, it was decided to use a grappling dummy to simulate falls. This dummy, which can be seen in Fig. 1, is of similar size and weight as an adult and was chosen to simulate the sounds of human falls during our data collection.

The data was captured using a setup that included a smartphone (Samsung Galaxy S23+) mounted on a tripod. Video clips of each fall were recorded with this configuration and recordings were performed in various scenarios, as detailed in Table 1.

Efforts were made to introduce randomness to improve the variability of fall sounds in the recordings. For falls from a standing position, techniques were varied: the dummy was either pushed to the ground to simulate a forceful fall or allowed to fall naturally, either forward or to the side. In addition, in some tests, the dummy was thrown against various objects such as walls, chairs, and desks. This method was periodically used to ensure that the dataset captured a wide range of fall-related sounds, including both simple falls and more complex interactions with different objects and surfaces.

The goal of the recording process was to collect at least 25 samples for each environment detailed in Table 1. However, the physical requirements for recording each sample influenced the number of samples collected for each scenario. Recording scenarios such as falls from stairs or from a lying position proved to be difficult. The main challenge came from the weight of the doll, approximately 140 pounds, which required considerable effort to repeatedly lift and place it on a table or other elevated platform before dropping it. Consequently, a smaller number of such samples were produced.

In contrast, recording falls from a standing position was comparatively less demanding, resulting in a larger number of these samples. Initially, samples were recorded in sets of 25. However, during data curation, some samples were discarded due to quality issues. Furthermore, in some cases, the number of samples recorded exceeded 25, causing discrepancies in the number of samples between different categories. Despite these variations, the collection ensured that at least 25 samples were taken for each type of fall, maintaining a consistent and representative dataset.

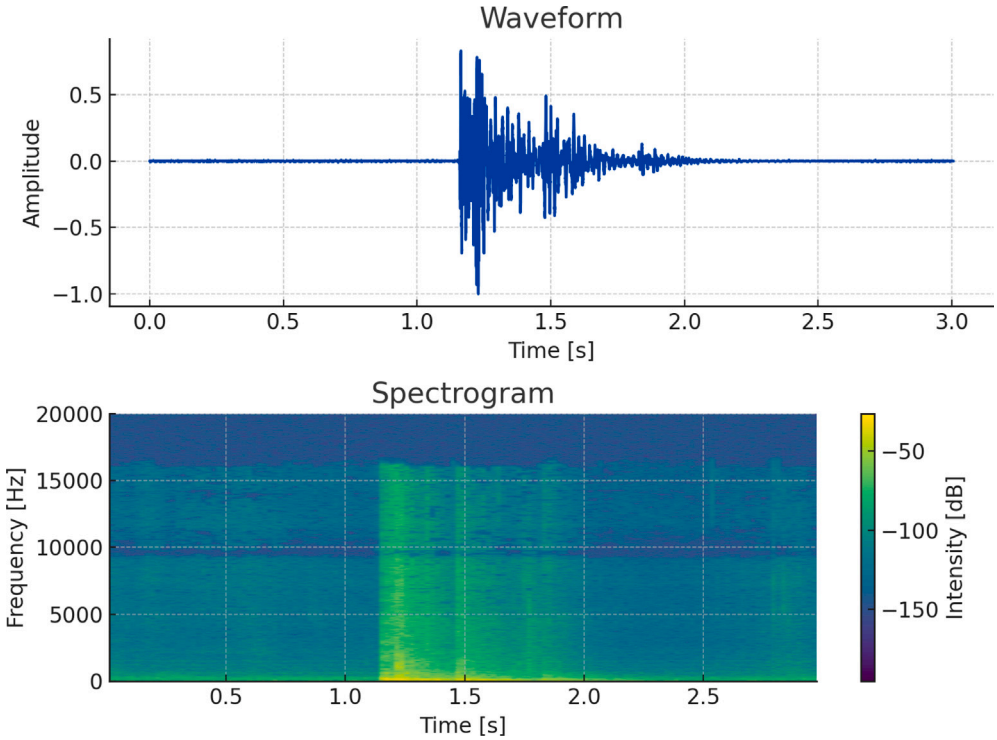

Fig. 2. Time and frequency representation of the original recording of a positive sample that represents the sound of a fall.

For the negative samples, audio data was recorded within a domestic setup with the intention of replicating background noise commonly encountered in a home environment where a fall detection system is targeted for deployment. This dataset includes various elements, including the sounds of individuals walking on both wood and carpet floors, typical kitchen noises such as dish cleaning and microwave operation, general cleaning sounds, and ambient background sounds from television and similar sources.

3.2. Data processing

After capturing instances of falls within groups of videos, a sequence of processing steps was applied. Initially, the audio signal from each video was extracted, followed by a smoothing process through averaging with a 50-sample running window. Subsequently, the signal was converted to its absolute value. Following this, an algorithm based on Hidden Markov Models (HMM) was used to identify sound spikes, serving as indicators of falls.

An analysis of the duration of the detected spikes was performed, indicating that a 3.0-second window would cover the majority of falling events. To isolate these events, the center of each audio spike was identified, and a 1.5 s segment was selected to the right and left of this center point. This approach ensured uniformity as all samples were of the same length.

Finally, a visual evaluation of these samples was performed, resulting in the identification of 475 instances that were categorized as “fall” events. The processing procedure described here is similar to the method described in the article named Huang and Garcia (2023). An example of a positive sample recorded can be seen in Fig. 2.

The audio recorded within domestic environments was partitioned into 3-second segments and subsequently shuffled to eliminate any potential correlation between consecutive samples. This meticulous process resulted in a total of 2410 negative samples. An example of a negative sample recorded can be seen in Fig. 3.

As can be seen in both images, positive and negative audio samples exhibit different visual features upon inspection. Positive samples show a prominent disturbance in the center, flanked by areas of minimal sound, indicating a clear, isolated event. In contrast, negative samples show a more uniform and random sound pattern, suggesting ongoing background activities.

It should be noted that positive samples were recorded in an environment with minimal background noise, as evidenced in Fig. 2. To make the dataset more practical, a strategy was employed to add background noise to the positive samples. This involved merging positive samples with randomly selected negative samples. Care was taken to ensure that each negative sample was used only once in the dataset since we have a large number of negative samples.

By mixing background noise with positive samples, the acoustic profiles of both types of samples became more closely aligned. This adjustment improves the applicability of the dataset for effective use in real-world fall detection. An example of a merged audio sample can be seen in Fig. 4.

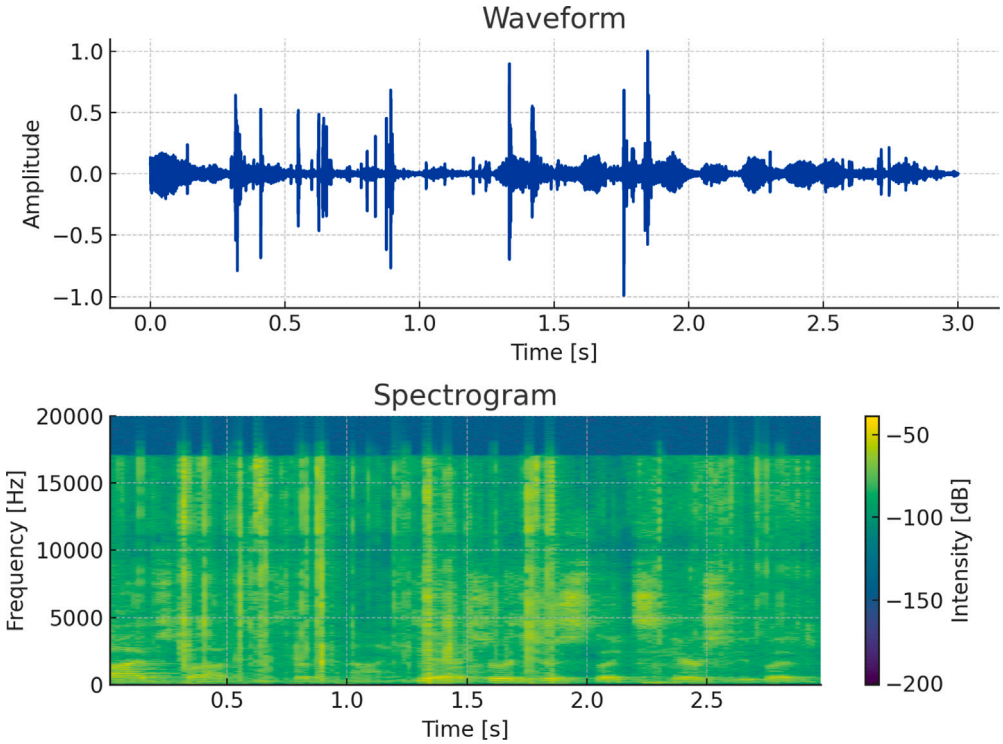

Fig. 3. Time and frequency representation of a randomly chosen negative sample without a fall event.

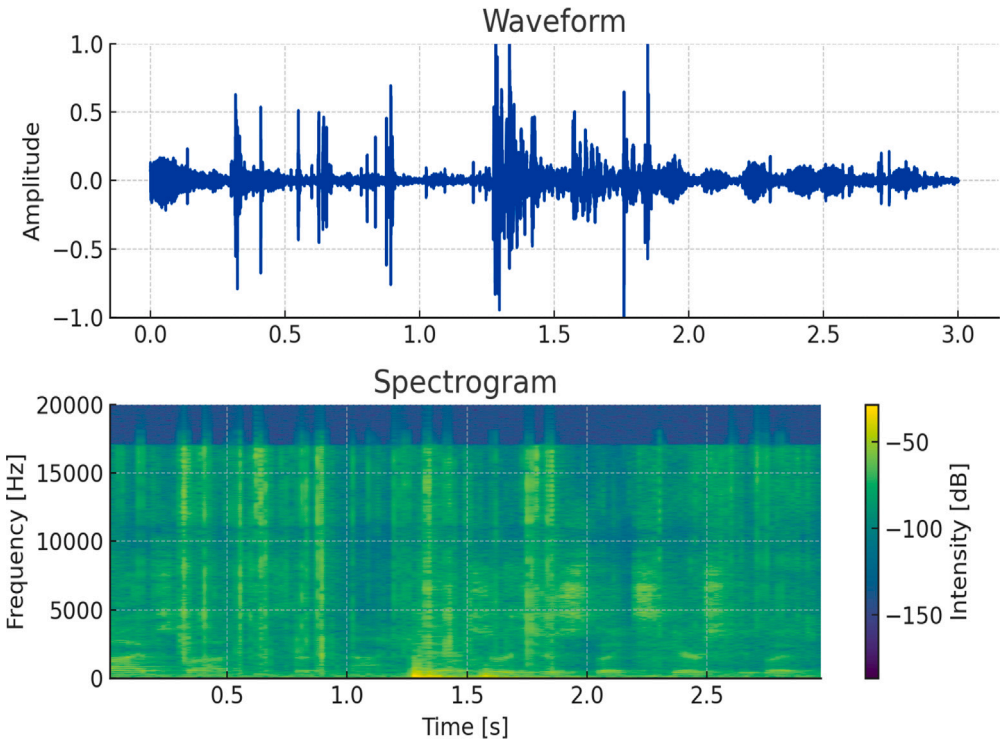

Fig. 4. Time and frequency representation of a randomly chosen positive sample from the dataset.

Table 2
Nomenclature for naming samples in the dataset.

Component	Description
AA	Fold number (01 to 10)
BBB	A numeric code randomly assigned for sample shuffling
CC	ID of the environment where the sample was recorded
DDD	Sequence number of the original video clip
FF	Class of the sample (e.g., 01 for fall, 02 for no fall)

3.3. Data segmentation

In the final phase of data curation for the fall detection dataset, a total of 950 samples were obtained, consisting of 475 positive and 475 negative samples. This dataset was partitioned into 10 folds to support the training and validation stages of machine learning.

A naming convention for the samples was also established to streamline the identification and organization of the data. The structure of this nomenclature, "AA-BBB-CC-DDD-FF", is detailed in Table 2.

The methodology detailed above resulted in the creation of the SAFE (Sound Analysis for Fall Events) dataset - a compilation of audio samples including fall and non-fall sounds. This dataset has been made publicly available in an online repository, allowing broader access and further research on audio-based fall detection techniques.

This dataset included 950 audio samples in WAV format at a sampling rate of 48 kHz. The 475 positive samples have already been added with background noise and the negative samples are original. The research findings and results presented in this paper are primarily based on analyzes conducted using this audio dataset. Furthermore, the dataset includes 475 original MP4 video clips, each capturing a different fall event. These clips serve as raw recorded data and offer opportunities for further processing, such as adding background noise using various techniques.

4. Performance evaluation of machine learning algorithms

This section describes the multiple experimental setups employed to evaluate the performance of various machine learning algorithms on the audio-based fall detection dataset. Three distinct approaches will be discussed, each designed to explore different aspects of feature extraction and classification techniques in the context of audio-based fall detection.

1. **Machine learning classification with digital audio samples**: The first approach involves performing a grid search to determine the most effective machine learning classification algorithms for this specific audio dataset. In this setup, digital audio samples are taken directly as input to the model.
2. **Machine learning classification with spectrogram-based feature extraction**: The second approach also employs a grid search to determine the best machine learning classifiers by transforming audio samples into spectrograms; thus, the classification task shifts toward image classification.
3. **Deep learning with spectrogram images**: The third approach involves the application of deep learning, specifically using a well-established image classification model. This model takes the input of spectrograms generated from the audio files.

Although a deep learning approach is anticipated to yield superior results due to its advanced pattern recognition capabilities, there is significant merit in exploring machine learning methods as well. These methods are notable for their simplicity and ease of implementation, particularly in the form of a low-power standalone embedded device. As mentioned earlier, creating a solution that can be effectively integrated into at-home systems is the key objective of this endeavor.

4.1. Machine learning classifiers for digital audio samples

In the literature, various machine learning algorithms, such as Support Vector Machines (SVM) (Adnan et al., 2018; Cheffena, 2016; Droghini, Principi et al., 2018; Hussain et al., 2019; Principi et al., 2016; Zhuang et al., 2009), Naive Bayes (Liu, Popescu, Rantz, & Skubic, 2012), k-Nearest Neighbors (k-NN) (Cheffena, 2016; Hussain et al., 2019), and Random Forests (Hussain et al., 2019), have been applied to audio datasets to develop classification algorithms. A notable trend across these studies is the reliance on spectrogram-based feature extraction methods to transform audio signals into more interpretable representations for classification. However, the possibility of directly using raw digital audio as input for machine learning models has not been widely explored.

In this study, raw audio data, in its digital form, is treated as the primary source of features for machine learning models. To evaluate the performance of various algorithms, a grid search was performed using Scikit-Learn's all_estimators tool, as previously detailed in Huang and Garcia (2023). This tool allowed for an extensive exploration of supervised learning classification algorithms, spanning Bayesian classifiers, decision trees, linear classifiers, nearest neighbors, support vector machines, probabilistic classifiers, and ensemble methods. For each algorithm, a 10-fold cross-validation was performed, ensuring robust performance evaluation by training and testing each model on multiple data partitions. Metrics such as accuracy, precision, recall and F1 score were calculated and averaged in the folds to assess algorithmic effectiveness.

One inherent challenge with using raw audio data is its high dimensionality. For example, audio sampled at 48 kHz for 3 s results in 144,000 individual data points per sample, which can increase computational demands and the risk of overfitting. However, preliminary tests with varying sample rates revealed an intriguing observation: reducing the audio resolution, thereby decreasing the number of input features, did not compromise classification accuracy. This finding suggests that not all data points in the raw format are critical for high classification performance. Strategic downsampling, therefore, offers a practical solution to reduce computational complexity while preserving or even enhancing model performance.

4.2. Machine learning with spectrogram features

In the field of audio classification, the use of spectrogram techniques for the extraction of features has been established as the predominant method. This approach facilitates the transformation of the challenge associated with audio classification into a problem similar to that associated with image classification. This conversion allows the application of techniques designed for image classification to audio data, thereby enhancing the likelihood of achieving accurate classification outcomes.

The focus on mel spectrograms has been particularly highlighted in the literature, as evidenced by various studies Cheffena (2016), Droghini et al. (2017), Droghini, Principi et al. (2018), Hussain et al. (2019), Li et al. (2012), Liu et al. (2012), Principi et al. (2016), Zigel et al. (2009), although they represent merely one of the available techniques. Research has also incorporated other methods for the extraction and analysis of audio features, including Short-Time Fourier Transform (STFT) (Huzaifah, 2017), Constant-Q Transform (CQT) Spectrograms (Dian Handy Permana & Bayu Yogha Bintoro, 2021), Wavelet Transforms (Waldekar & Saha, 2018), Chroma Features (Shi, Li, & Tian, 2019), and Mel-Frequency Cepstral Coefficients (MFCCs) (Khunarsal, Lursinsap, & Raicharoen, 2013). These methodologies offer a variety of approaches to capture the fundamental features of audio signals, including aspects such as spectral and temporal features, as well as harmonic and melodic content. Examples of these spectrogram techniques, applied to a single audio file, are illustrated in Fig. 5.

The grid search methodology to be applied in this context adheres to the protocol delineated in the preceding section, implementing the all_estimators tool as provided by Scikit-Learn. The computation of spectrograms will be conducted on the audio samples contained within the dataset, specifically employing the 48 kHz samples.

Despite not being the state of the art in image classification, the machine learning algorithms being evaluated are still valuable, particularly for edge computing devices. This approach aims to evaluate the accuracy achievable with basic algorithms, intentionally excluding deep learning techniques from this specific application.

4.3. Deep learning with spectrogram images

The dataset was evaluated using three state-of-the-art image classification models: ResNet-18, EfficientNet-B0, and ConvNeXt-Small. These models were chosen because of their proven performance, relatively small size, and manageable number of parameters, ensuring that the evaluation remained fair and impartial without significant differences in model complexity or capacity. Furthermore, these models were selected based on their previous use in the literature for sound classification tasks, where they have exhibited effective feature extraction and classification capabilities in similar domains. Using deep learning algorithms to classify spectrograms derived from audio samples is a standard approach in sound classification research (Incze, Jancsó, Szilágyi, Farkas, & Sulyok, 2018; Li, Wang, Wang, Qiao, & Zhou, 2022; Pellegrini, Khalfaoui-Hassani, Labbé, & Masquelier, 2023; Haq et al., 2021; Zakaria, Mohamed, Abdelghani, & Sundaraj, 2021).

Pre-trained versions of these models, available through the PyTorch framework, were used to simplify implementation. PyTorch provides tools to download pre-trained weights on large datasets such as ImageNet, enabling the use of transfer learning. This approach reduces the computational resources required for training while allowing the models to benefit from features learned in diverse datasets.

Simpler configurations of these models were intentionally selected to match the goals of this project. In potential deployment scenarios, particularly those that involve embedded hardware, it would be essential to use models that are lightweight and efficient. Embedded systems often come with strict limitations in memory, processing power, and energy consumption, making it necessary to optimize the models for these constraints. These models must strike a balance between computational efficiency and the ability to deliver better performance when applied to tasks like fall detection. By prioritizing models that are compatible with embedded hardware, the framework can be adapted to applications in resource-constrained environments while maintaining practical utility.

5. Results

This section presents the results of the evaluations conducted on the dataset, as described in the experimental setups described above. For clarity, the metrics used to evaluate the algorithms are as follows.

- **Accuracy:** measures the proportion of true results (both true positives and true negatives) among the total number of cases examined. It can be calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where TP is True Positive, TN is True Negative, FP is False Positive, and FN is False Negative.

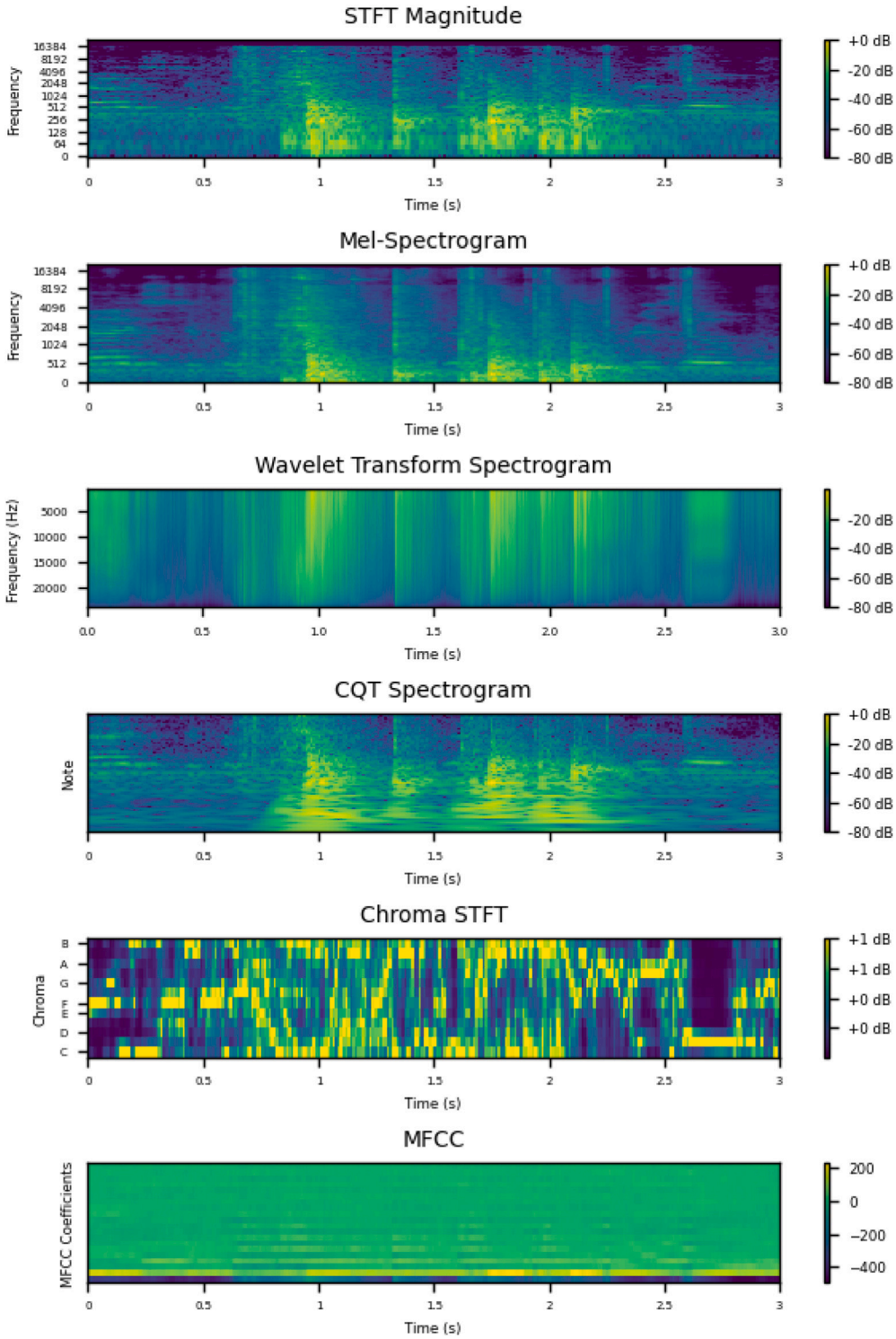

Fig. 5. This figure displays six spectrogram visualizations derived from a single audio file. It includes the STFT Magnitude Spectrogram, Mel-Spectrogram, Wavelet Transform Spectrogram, CQT (Constant-Q Transform) Spectrogram, Chroma STFT, and MFCC (Mel-Frequency Cepstral Coefficients).

- **Precision**: (or Positive Predictive Value) measures the proportion of true positives among all positive results (both true positives and false positives). It is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** (or Sensitivity, True Positive Rate) measures the proportion of true positives identified among all actual positives. It is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score:** is the harmonic mean of Precision and Recall. It is given by:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The chosen evaluation metrics are essential to evaluate the performance of the algorithms in the SAFE dataset. Accuracy provides a simple measure of the overall performance of the model, indicating the proportion of true positive and negative predictions. Precision and recall provide information about the model's ability to correctly predict fall events and minimize false positives, crucial for fall detection systems where precision and reliability are paramount. The F1 score combines precision and recall, offering a unique metric that balances the trade-offs between the two. This set of metrics ensures a complete evaluation of the effectiveness and accuracy of the model in fall detection.

5.1. Machine learning algorithms on digital audio samples

In this analysis, the Scikit-Learn all_estimators tool was used to assess 30 distinct classification algorithms. Using a 10-fold cross-validation approach, each algorithm was evaluated with 9 folds designated for training and 1 fold for testing purposes. Throughout each training phase, key performance metrics were recorded. Upon completion of the cross-validation process, the average scores for these metrics were computed to provide an overview of each algorithm's performance.

This process was executed on an audio dataset that included the original samples recorded at 48 kHz and their resampled versions at 32 kHz, 16 kHz, 8 kHz, and 4 kHz. The objective was to understand how the fidelity of audio samples affects the accuracy and effectiveness of different machine learning models in classifying audio data. The results of this analysis are summarized in Table 3.

As can be seen in the results, the decision tree-based algorithms consistently emerge as the best performers in each evaluated case. Specifically, ensemble methods such as the ExtraTreesClassifier and GradientBoostingClassifier demonstrated their effectiveness in leveraging multiple decision trees to achieve superior performance.

The ExtraTreesClassifier and the ExtraTreeClassifier, despite their similar names, differ significantly in their approach to decision tree construction. ExtraTreesClassifier is an ensemble method that builds multiple randomized decision trees and aggregates their predictions, leading to more accurate and stable results. However, the ExtraTreeClassifier represents a single, extremely randomized decision tree that lacks the ensemble advantage that the ExtraTreesClassifier benefits from. This distinction is crucial, as ExtraTreesClassifier generally outperforms ExtraTreeClassifier due to its ensemble nature, similar to how Random Forest operates by creating a forest of trees, but with less randomness in split selection compared to ExtraTreesClassifier.

Probabilistic methods such as Gaussian Naive Bayes also exhibit very strong performance, particularly in terms of recall. This high recall rate is a clear indication of Gaussian Naive Bayes' ability to correctly identify a high proportion of true positive cases, which in the context of audio-based fall detection systems, translates to accurately detecting fall events. Given the critical nature of these systems, where missing a true fall event could have serious consequences, the high recall accuracy of probabilistic methods becomes particularly valuable.

5.2. Machine learning algorithms on spectrogram features

For this analysis, each audio sample was transformed into a PNG image file using six different spectrogram techniques previously described.

To align with the input format favored by various deep learning algorithms, images were exported in grayscale with a resolution of 224x224 pixels. This standardized format makes it easier to evaluate various models. Consequently, each image was processed into a flattened vector comprising 50,176 features, preparing the data for the application of the machine learning algorithm.

The results from the grid-searched 10-fold cross-validation are summarized in Table 4.

As evidenced by the results, there are four spectrogram techniques—Mel Spectrogram, STFT Spectrogram, MFCC Spectrogram, and Constant Q Transform Spectrogram that consistently exhibit similar high performance across various classification algorithms.

Across the four high-performing spectrogram techniques, the top-performing algorithms are predominantly linear models. This includes Linear Discriminant Analysis, Ridge Classifier, Logistic Regression, and LinearSVC, which consistently achieve high metrics in accuracy, precision, recall, and F1-score across these spectrogram methods.

The success of these linear models in different spectrogram representations indicates a significant level of linear separability within the feature spaces they create. This observation is important because it highlights the inherent features of the audio data represented by these spectrogram techniques. The prominence of linear models as top performers suggests that these methods

Table 3
Performance comparison of various classification algorithms applied to the SAFE dataset across different audio sampling rates (48 kHz, 32 kHz, 16 kHz, 8 kHz, and 4 kHz). Metrics reported include Accuracy, Precision, Recall, and F1-score, evaluated using a 10-fold cross-validation method. Decision tree-based ensemble models, such as ExtraTreesClassifier and GradientBoostingClassifier, consistently exhibit high performance across all metrics and sampling rates, achieving accuracy above 93% even at reduced sampling rates. GaussianNB achieves the highest Recall, indicating robust detection of positive cases, but with slightly lower overall accuracy. These results suggest that essential features for classification are preserved even at lower sampling rates, enabling the development of computationally efficient, resource-constrained fall detection systems.

Classifier algorithm	Accuracy	Precision	Recall	F1-score
Results obtained with 48 kHz sampling rate				
GradientBoostingClassifier	91.68%	92.42%	90.97%	91.57%
ExtraTreesClassifier	91.58%	90.57%	92.89%	91.66%
HistGradientBoostingClassifier	91.26%	90.03%	92.89%	91.37%
GaussianNB	90.63%	85.50%	98.10%	91.31%
RandomForestClassifier	90.53%	88.98%	92.66%	90.70%
SVC	90.11%	89.32%	91.10%	90.17%
NuSVC	89.79%	87.58%	92.82%	90.10%
BaggingClassifier	88.74%	86.35%	92.19%	89.09%
AdaBoostClassifier	88.00%	88.41%	87.56%	87.90%
ExtraTreeClassifier	79.79%	81.01%	77.98%	79.31%
Results obtained with 32 kHz sampling rate				
ExtraTreesClassifier	91.79%	91.01%	92.87%	91.85%
GradientBoostingClassifier	91.58%	91.69%	91.59%	91.54%
HistGradientBoostingClassifier	91.47%	90.55%	92.66%	91.52%
RandomForestClassifier	90.84%	89.49%	92.66%	90.99%
GaussianNB	90.74%	85.41%	98.52%	91.44%
SVC	90.53%	89.57%	91.73%	90.61%
NuSVC	90.21%	88.46%	92.61%	90.46%
AdaBoostClassifier	88.11%	87.59%	88.89%	88.16%
BaggingClassifier	87.89%	84.74%	92.64%	88.44%
ExtraTreeClassifier	80.32%	81.58%	78.30%	79.83%
Results obtained with 16 kHz sampling rate				
ExtraTreesClassifier	92.42%	91.61%	93.49%	92.48%
GradientBoostingClassifier	91.79%	91.49%	92.24%	91.80%
GaussianNB	91.16%	85.92%	98.73%	91.82%
RandomForestClassifier	91.16%	90.00%	92.66%	91.26%
HistGradientBoostingClassifier	90.84%	89.58%	92.40%	90.94%
NuSVC	90.53%	88.54%	93.24%	90.79%
SVC	90.53%	89.71%	91.52%	90.59%
BaggingClassifier	88.74%	86.36%	91.99%	89.06%
AdaBoostClassifier	87.68%	87.62%	88.00%	87.66%
ExtraTreeClassifier	82.00%	83.22%	80.27%	81.63%
Results obtained with 8 kHz sampling rate				
ExtraTreesClassifier	92.63%	91.59%	93.91%	92.71%
GradientBoostingClassifier	92.53%	92.44%	92.68%	92.51%
HistGradientBoostingClassifier	92.11%	91.38%	93.08%	92.16%
RandomForestClassifier	91.37%	89.74%	93.49%	91.50%
GaussianNB	91.16%	85.75%	99.14%	91.87%
SVC	90.63%	89.45%	92.19%	90.77%
NuSVC	90.42%	88.52%	93.03%	90.68%
AdaBoostClassifier	89.79%	90.29%	89.25%	89.71%
BaggingClassifier	88.00%	85.21%	92.01%	88.44%
ExtraTreeClassifier	82.21%	83.28%	80.54%	81.80%
Results obtained with 4 kHz sampling rate				
ExtraTreesClassifier	93.26%	92.96%	93.73%	93.26%
GradientBoostingClassifier	92.11%	92.05%	92.22%	92.07%
HistGradientBoostingClassifier	91.79%	91.42%	92.24%	91.79%
RandomForestClassifier	91.05%	89.22%	93.49%	91.22%
GaussianNB	91.37%	85.86%	99.35%	92.05%
SVC	91.26%	90.54%	92.19%	91.33%
NuSVC	90.84%	89.25%	93.03%	91.05%
AdaBoostClassifier	90.53%	90.04%	91.18%	90.58%
BaggingClassifier	89.16%	87.66%	91.39%	89.41%
ExtraTreeClassifier	83.89%	75.84%	77.98%	76.17%

effectively transform complex audio signals into feature spaces that are largely linearly separable. Linear models are useful for their simplicity, interpretability, and computational efficiency, which are advantageous characteristics for analyzing large datasets.

Table 4
Performance comparison of classification algorithms applied to the SAFE dataset using various spectrogram-based feature extraction methods, including Mel Spectrogram, STFT Spectrogram, MFCC Spectrogram, Constant Q Transform Spectrogram, Continuous Wavelet Transform Spectrogram, and Chroma Features Spectrogram. Metrics reported are Accuracy, Precision, Recall, and F1-score, evaluated using a 10-fold cross-validation process. Linear models, such as Linear Discriminant Analysis (LDA) and Ridge Classifier, achieve the highest performance across most spectrogram methods, with Accuracy exceeding 97% for Mel and Constant Q Transform Spectrograms. The Ridge Classifier exhibits consistent robustness, making it suitable for lightweight deployment. Conversely, performance drops slightly for Continuous Wavelet Transform and Chroma Features Spectrograms, indicating less effective feature representation.

Classifier algorithm	Accuracy	Precision	Recall	F1-score
Results obtained with Mel Spectrogram				
LinearDiscriminantAnalysis	97.37%	97.41%	97.38%	97.36%
RidgeClassifier	97.16%	97.18%	97.18%	97.16%
LinearSVC	96.32%	96.34%	96.33%	96.31%
LogisticRegression	96.21%	96.24%	96.22%	96.21%
GradientBoostingClassifier	95.89%	95.95%	95.92%	95.89%
RandomForestClassifier	95.89%	95.92%	95.94%	95.89%
HistGradientBoostingClassifier	95.89%	95.93%	95.93%	95.89%
PassiveAggressiveClassifier	95.58%	95.65%	95.59%	95.57%
Results obtained with STFT Spectrogram				
RidgeClassifier	97.05%	97.10%	97.08%	97.05%
PassiveAggressiveClassifier	96.95%	97.02%	96.98%	96.94%
LogisticRegression	96.95%	97.00%	96.98%	96.94%
SVC	96.84%	96.88%	96.87%	96.84%
LinearDiscriminantAnalysis	96.84%	96.89%	96.85%	96.84%
LinearSVC	96.74%	96.78%	96.76%	96.73%
ExtraTreesClassifier	96.11%	96.13%	96.12%	96.10%
Perceptron	96.00%	96.09%	96.02%	96.00%
Results obtained with MFCC Spectrogram				
LinearDiscriminantAnalysis	96.53%	96.66%	96.58%	96.52%
RidgeClassifier	96.00%	96.04%	96.05%	96.00%
LogisticRegression	95.89%	95.97%	95.92%	95.89%
LinearSVC	95.79%	96.00%	95.85%	95.78%
HistGradientBoostingClassifier	94.21%	94.31%	94.24%	94.20%
SVC	94.11%	94.17%	94.12%	94.10%
RandomForestClassifier	93.37%	93.47%	93.41%	93.36%
ExtraTreesClassifier	93.16%	93.27%	93.22%	93.15%
Results obtained with Constant Q Transform Spectrogram				
RidgeClassifier	97.47%	97.52%	97.49%	97.47%
LinearDiscriminantAnalysis	97.47%	97.51%	97.51%	97.47%
LinearSVC	97.37%	97.41%	97.39%	97.36%
LogisticRegression	97.05%	97.11%	97.09%	97.05%
PassiveAggressiveClassifier	96.84%	96.91%	96.88%	96.84%
SGDClassifier	96.74%	96.81%	96.79%	96.73%
SVC	96.63%	96.70%	96.66%	96.63%
ExtraTreesClassifier	96.42%	96.46%	96.45%	96.42%
Results obtained with Continuous Wavelet Transform Spectrogram				
HistGradientBoostingClassifier	88.11%	88.31%	88.15%	88.09%
SVC	87.26%	87.48%	87.27%	87.24%
ExtraTreesClassifier	86.53%	86.65%	86.55%	86.51%
RandomForestClassifier	86.42%	86.62%	86.42%	86.39%
GradientBoostingClassifier	85.89%	86.13%	85.93%	85.87%
AdaBoostClassifier	85.58%	85.80%	85.57%	85.54%
NuSVC	85.26%	85.50%	85.27%	85.22%
BaggingClassifier	84.11%	84.26%	84.11%	84.07%
Results obtained with Chroma Features Spectrogram				
SVC	86.84%	87.34%	86.90%	86.80%
NuSVC	86.74%	87.30%	86.79%	86.69%
GradientBoostingClassifier	86.21%	86.55%	86.29%	86.18%
HistGradientBoostingClassifier	86.11%	86.42%	86.11%	86.07%
RandomForestClassifier	84.74%	85.47%	84.76%	84.65%
ExtraTreesClassifier	84.11%	84.59%	84.15%	84.05%
KNeighborsClassifier	83.58%	83.74%	83.60%	83.55%
AdaBoostClassifier	82.21%	82.35%	82.22%	82.18%

In addition to the observed effectiveness of linear models, decision trees and support vector machines (SVC) showed good performance. These models were evaluated using their default hyperparameters. Consequently, it is possible that the performance metrics for each model could be further optimized through hyperparameter tuning, indicating the potential for better results.

Table 5
Performance comparison of deep learning models (ConvNeXt Small, ResNet18, and EfficientNet B0) applied to the SAFE dataset using different spectrogram-based feature extraction methods, including Mel Spectrogram, STFT Spectrogram, MFCC Spectrogram, Constant Q Transform Spectrogram, Continuous Wavelet Transform Spectrogram, and Chroma Features Spectrogram. Metrics reported are Accuracy, Precision, Recall, and F1-score, evaluated via a 10-fold cross-validation approach. ConvNeXt Small consistently achieves the highest performance across most spectrogram methods, with a maximum accuracy of 99% using the STFT Spectrogram. ResNet18 also achieved strong performance but slightly lags behind ConvNeXt Small, while EfficientNet B0 achieves moderate results, requiring more epochs for optimization. Performance diminishes for Continuous Wavelet Transform and Chroma Features Spectrograms, reflecting their less effective representation of fall detection features. These results emphasize the critical role of both spectrogram method and model selection in achieving optimal fall detection accuracy.

Classifier algorithm	Accuracy	Precision	Recall	F1-score
Results obtained with Mel Spectrogram				
ConvNeXt Small	98.95%	99.37%	98.51%	98.93%
ResNet18	97.68%	98.31%	97.04%	97.67%
EfficientNet B0	94.63%	96.40%	93.27%	94.62%
Results obtained with STFT Spectrogram				
ConvNeXt Small	99.05%	99.79%	98.38%	99.06%
ResNet18	98.32%	99.16%	97.49%	98.29%
EfficientNet B0	96.42%	98.50%	94.38%	96.34%
Results obtained with MFCC Spectrogram				
ConvNeXt Small	96.11%	98.11%	94.12%	96.03%
EfficientNet B0	89.89%	90.02%	91.21%	90.07%
ResNet18	86.84%	89.86%	89.84%	88.41%
Results obtained with Constant Q Transform Spectrogram				
ConvNeXt Small	97.89%	98.75%	97.08%	97.88%
ResNet18	97.05%	97.93%	96.27%	97.04%
EfficientNet B0	95.05%	98.47%	91.63%	94.88%
Results obtained with Continuous Wavelet Transform Spectrogram				
ConvNeXt Small	89.58%	92.25%	86.53%	89.17%
EfficientNet B0	85.58%	85.74%	86.71%	85.76%
ResNet18	83.79%	82.85%	85.50%	84.07%
Results obtained with Chroma Features Spectrogram				
ConvNeXt Small	90.53%	91.79%	89.33%	90.45%
EfficientNet B0	86.00%	88.98%	82.48%	85.46%
ResNet18	85.79%	85.55%	86.20%	85.76%

5.3. Deep learning algorithms on spectrogram images

Training and evaluation of the three algorithms, ResNet-18, EfficientNet-B0, and ConvNeXt-Small, were carried out over 100 epochs, with all models sharing the same learning rate for consistency. The dataset was subjected to a 10-fold cross-validation process, where, in each iteration, 9 folds were combined and then split in an 80:20 ratio for training and validation, respectively. The 10th fold was reserved exclusively for testing. This setup ensured that each segment of the dataset was used as a test set exactly once, allowing for a comprehensive evaluation of the model performance.

For each cross-validation fold, the models were trained from the ground up, starting with no pre-learned weights from any previous folds. This strategy was adopted to maintain the integrity of the evaluation, ensuring that the performance metrics obtained were a result of learning from the current fold's training data only.

The results, which are the average of the results from the 10 folds used as test datasets, are detailed in Table 5.

This setup ensured that each part of the dataset was tested once, providing a clear picture of how well the models performed. Deep learning models achieved superior results compared to traditional machine learning models on all types of spectrograms, especially ConvNeXt, which achieved over 99% accuracy with the STFT spectrogram. ConvNeXt stood out as the best performer in all cases, compared to ResNet and EfficientNet. However, it is important to note that this model is more complex and requires more training time than the others.

A constant training duration of 100 epochs was chosen to maintain consistency, but it was observed that both ResNet and ConvNeXt could achieve satisfactory results in 50 or even 25 epochs. EfficientNet, on the other hand, required the full 100 epochs for this task, indicating the need for more training. Given the cross-validation process, where one fold was always kept as unseen data, concerns about overfitting were mitigated, ensuring that the performance of the models genuinely reflected their ability to generalize to new data.

6. Discussion and future work

An extensive evaluation of various machine learning techniques was conducted on digitized audio samples, with different feature extraction methods applied. Significant results were found to be achieved using algorithms such as the ExtraTreesClassifier, a decision tree-based method, which achieved a maximum accuracy of more than 93% with just 4 kHz samples.

Interestingly, it was observed that reducing the sampling rate from 48 kHz to 4 kHz led to slightly improved performance of machine learning algorithms. This finding implies that lower sampling rates do not necessarily compromise the ability of algorithms to effectively classify audio samples. It suggests that the critical features necessary for classification are preserved even when the resolution of the data is decreased. This can result in significant reductions in computational demands and processing times.

This aspect is particularly crucial for projects focused on implementation, where privacy concerns are paramount. For instance, in the development of a fall detection device, the goal is often to operate these models locally, either on microcontrollers or embedded processors. This local processing minimizes the need to transmit data over the cloud, addressing privacy concerns associated with some audio-based systems. Being able to run a classic model like a decision tree on a device with low power consumption is immensely valuable. It ensures that essential services, such as fall detection, can be provided efficiently and privately by processing data directly on the device, thereby enhancing the practicality and security of deploying these technologies in sensitive environments.

By evaluating multiple machine learning models through a grid search approach combined with a 10-fold cross-validation process, it was possible to identify the best performers for this task, which turned out to be decision trees. However, some intriguing results were observed, notably with the Gaussian Naive Bayes classifier. Despite having a lower accuracy compared to decision tree models, the Gaussian Naive Bayes classifier achieved a high recall, reaching up to 99.14% with the 4 kHz samples. This indicates that the model is very precise in detecting the class representing true positives, such as actual falls in a fall detection context. This ability to accurately identify true positive events with a relatively low number of features signifies that the Gaussian Naive Bayes classifier could play a critical role in an ensemble method.

Such an ensemble method could take advantage of the strong performance of the Naive Bayes Gaussian classifier to reliably detect true positive events. Meanwhile, the evaluation of the other class could be improved by a bootstrap combination of other high-performance algorithms. This strategy would ensure a robust system that maximizes the detection accuracy of both classes, utilizing the strengths of multiple algorithms. The potential to run these models on devices with low power consumption and sampling rates highlights the importance of this approach.

The use of spectrograms as a feature extraction technique presented improved performance, with a notable achievement of 97.47% precision using a Ridge classifier on a constant Q-transform spectrogram. This finding is particularly interesting given the predominant preference for Mel spectrograms in the existing literature for feature extraction. However, the results of this study indicate that a wider range of machine learning algorithms showed better results with the constant Q transform spectrogram compared to the Mel spectrogram. Although the difference in performance of the best performing model is not significant, it is noted that each of the best models performed better with this specific spectrogram technique.

This observation may not be true for all audio classification tasks, suggesting the importance of evaluating multiple spectrogram techniques on multiple audio datasets to determine the most universally effective feature extraction method. Interestingly, the deep learning results showed superior performance with the short-time Fourier transform (STFT) spectrogram. It is also worth noting that each spectrogram technique comes with its own set of hyperparameters, pointing to the possible need for optimization evaluations to identify the best settings for feature extraction. This area of research could be crucial for future studies on audio classification, with the goal of refining and optimizing the process of extracting meaningful features from audio signals to further improve model performance.

Regardless of the spectrogram technique employed, linear models emerged as the more dominant performers in this evaluation, with the Ridge Classifier being at the top or near the top across 4 of the 6 evaluated spectrograms. This is a significant observation, especially considering the characteristics of the Ridge Classifier. As a linear model, the Ridge Classifier is notably easy to implement on a microcontroller due to its light computational demand and low power consumption. For the reasons mentioned above, the capability of linear models to effectively address this task is highly valued, particularly with the prospect of hardware implementation of such algorithms.

As expected, deep learning models exhibited the best performance in this study, primarily because the audio classification problem was transformed into the image classification domain, where state-of-the-art methods have been refined and optimized. The top performer was ConvNeXt, with 99.05% classification accuracy using STFT for feature extraction.

However, despite the superior performance of deep learning models, the machine learning algorithms were so close to the top performers that it can be argued that for this specific task, deep learning is preferred but not essential to achieve high performance. Especially noteworthy is the potential to leverage models like the previously mentioned Gaussian Naive Bayes and to explore bootstrapping multiple models. This strategy aims to maximize performance while maintaining the simplicity and lightweight nature of the models.

For future endeavors, the implementation of these algorithms in hardware prototypes will be pursued with the aim of developing a standalone solution that could be used in conjunction with other well-established techniques for fall detection. This is particularly expected to benefit the elderly population, offering a reliable and efficient method for falling detection to improve safety and improve response times early.

Improvements should be sought by further exploration into the audio classification field, using strategies mentioned above to enhance classification accuracy through the application of multiple feature extraction techniques. This involves harnessing the strengths of certain machine learning models that have shown significant potential in the developed dataset.

7. Limitations of the study

Despite advancements in audio-based fall detection, several limitations must be acknowledged to contextualize the findings and guide future work:

1. Dataset Constraints: The SAFE dataset, while novel, was created using a grappling dummy to simulate falls. Although the dummy was chosen to approximate the human size and weight, its acoustic properties may differ from the acoustic properties of real human falls. This discrepancy could affect the generalizability of the models to real-world scenarios involving human subjects.
2. Controlled Environment: Data collection was performed in controlled environments with minimal background noise for positive samples. Although steps were taken to merge fall sounds with background noise from negative samples, this approach may not fully replicate the complex and dynamic noise environments encountered in real-world settings, such as busy households or care facilities.
3. Limited Real-World Testing: The models were evaluated using curated datasets under controlled experimental conditions. The robustness of the system in detecting falls in diverse real-world scenarios, including those that involve multiple overlapping sound sources, remains untested.
4. Impact of Sampling Rates: Although the study found that reducing the sampling rate did not adversely affect the performance of the model, more research is needed to determine the minimum resolution necessary for consistent performance under varying acoustic conditions.
5. Model Complexity vs. Deployment Feasibility: Although deep learning models achieved superior accuracy, their higher computational requirements may limit the deployment of embedded devices with limited resources. Although simpler machine learning models were evaluated, a more detailed analysis of computational trade-offs, including latency and power consumption, is necessary for practical implementation.
6. Ethical and Privacy Considerations: The use of audio-based detection raises potential concerns about privacy. The extent to which the system can ensure user privacy while maintaining performance, especially in sensitive environments, needs further exploration.
7. Generalizability to Different Populations: The dataset primarily simulated adult fall scenarios. The system's performance for other populations, such as children or individuals with mobility aids (e.g., wheelchairs or walkers), remains unexplored.

Addressing these limitations will be crucial to advance the applicability and reliability of audio-based fall detection systems. Future research should focus on expanding the dataset, incorporating real-world trials, and exploring innovative privacy-preserving approaches to ensure broader adoption of this technology in diverse environments.

8. Conclusion

In this study, we built a SAFE (Sound Analysis for Fall Events) dataset as a unique compilation of 950 audio samples, including 475 fall events recorded in multiple configurations using a grappling dummy. This dataset has been curated and released to the public, standing out as one of the few datasets available in the field of audio-based fall detection. The dataset was extensively evaluated, employing a variety of machine learning and deep learning algorithms along with various feature extraction methods, such as multiple spectrograms and direct use of the digitized audio as input to the models.

This evaluation provides the performance of different machine learning techniques on the task of audio-based fall detection. In particular, decision tree models directly using digitized audio samples as input achieved accuracy greater than 93%. Furthermore, the Naive Bayes Gaussian classifier reached a high recall rate of 99%, suggesting potential avenues for the development of ensemble methods that take advantage of this strength.

When using spectrograms for feature extraction, linear models, particularly the Ridge Classifier, showed an accuracy that exceeded 97%. Meanwhile, among deep learning algorithms, ConvNeXt stood out with a maximum accuracy of 99%. This extensive analysis not only underscores the viability of various computational approaches for audio-based fall detection, but also highlights the potential for further optimizing these methods, offering promising directions for future research in enhancing fall detection systems for the elderly and other at-risk populations.

CRediT authorship contribution statement

Antony Garcia: Investigation, Data curation, Conceptualization. **Xinming Huang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset for Sound Analysis for Fall Events (SAFE) used in this research can be accessed publicly on Kaggle via the following URL: https://www.kaggle.com/dsv/7516242. This dataset comprises the audio files used in this study as well as the original video recording from which these audio files were derived.

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